

Process capability ratio related problems

Problem-1

The specifications for one characteristic of a part call for its width to be 3.000 ± 0.008 cm. The process has been run under controlled conditions so that no assignable causes of variation have been introduced, and samples have been taken. The standard deviation of the process was estimated to be 0.003 cm.

- Cp What is Cp for this process? What does this value say about the process capability?
- What percentage of the units would be outside the specifications for the width of the part if the process were operated and centered at 3.000 centimeters? Assume a normal distribution of measurements.

Solution:

- a) Given, Target width=3.000 cm

$$USL=3000+0.008= 3.008 \text{ cm}$$

$$LSL= 3000-0.008=2.992 \text{ cm}$$

$$\text{Standard deviation, } \sigma = 0.003 \text{ cm}$$

Process capability index, $C_p = \frac{USL - LSL}{6\sigma} = (3.008 - 2.992) / (6 \times 0.003) = 0.8889 < 1$; the process is not capable of consistently producing parts within the specification limits.

- b) For USL 3.008 cm, Target width =3.000 cm

$$\text{So, percentage of units for the outside specification of USL} = (3.000 \div 3.008) \times 100 = 99.73\%$$

$$\text{\& Percentage of units for the outside specification of USL} = (3.000 \div 2.992) \times 100 = 100.27\%$$

Problem-2

<i>Subgroup Number</i>	<i>Date</i>	<i>Time</i>	<i>X₁</i>	<i>X₂</i>	<i>X₃</i>	<i>X₄</i>
<i>1</i>	<i>23-Jul</i>	<i>8:50</i>	<i>35</i>	<i>40</i>	<i>32</i>	<i>37</i>
<i>2</i>		<i>11:30</i>	<i>46</i>	<i>37</i>	<i>36</i>	<i>41</i>
<i>3</i>		<i>1:45</i>	<i>34</i>	<i>40</i>	<i>34</i>	<i>36</i>
<i>4</i>		<i>3:45</i>	<i>69</i>	<i>64</i>	<i>68</i>	<i>59</i>
<i>5</i>		<i>4:20</i>	<i>38</i>	<i>34</i>	<i>44</i>	<i>40</i>
<i>6</i>	<i>29-Jul</i>	<i>9:25</i>	<i>41</i>	<i>40</i>	<i>29</i>	<i>34</i>
<i>7</i>		<i>11:00</i>	<i>38</i>	<i>44</i>	<i>28</i>	<i>58</i>
<i>8</i>		<i>2:35</i>	<i>35</i>	<i>41</i>	<i>37</i>	<i>38</i>
<i>9</i>		<i>3:15</i>	<i>56</i>	<i>55</i>	<i>45</i>	<i>48</i>
<i>10</i>	<i>30-Jul</i>	<i>9:35</i>	<i>38</i>	<i>40</i>	<i>45</i>	<i>37</i>
<i>11</i>		<i>10:20</i>	<i>39</i>	<i>42</i>	<i>35</i>	<i>40</i>
<i>12</i>		<i>11:35</i>	<i>42</i>	<i>39</i>	<i>39</i>	<i>36</i>
<i>13</i>		<i>2:00</i>	<i>43</i>	<i>36</i>	<i>35</i>	<i>38</i>
<i>14</i>		<i>4:25</i>	<i>39</i>	<i>38</i>	<i>43</i>	<i>44</i>

Compute the C_p and C_{pk} from the following data table

Solution:

Subgroup Number	X1	X2	X3	X4	Average ($\bar{X}$)	Range (R)
1	35	40	32	37	36	8
2	46	37	36	41	40	10
3	34	40	34	36	36	6
4	69	64	68	59	65	10
5	38	34	44	40	39	10
6	41	40	29	34	36	12
7	38	44	28	58	42	30
8	35	41	37	38	37.75	6
9	56	55	45	48	51	11
10	38	40	45	37	40	8
11	39	42	35	40	39	7
12	42	39	39	36	39	6
13	43	36	35	38	38	8
14	39	38	43	44	41	6
Xbar					41.41071429	
R bar					9.857142857	

Xbar			R bar		
UCL	CL	LCL	UCL	CL	LCL
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0

For calculating C_p and C_{pk} ,

We know,

Process Capability Ratio (PCR) = $C_p = (\text{USL} - \text{LSL}) / 6\sigma$

$$C_{pk} = \frac{\text{Min} \{(\text{USL} - \bar{X}) \text{ or } (\bar{X} - \text{LSL})\}}{3\sigma}$$

$$\sigma = \bar{R}/d_2$$

Sample Size	CHART FOR RANGES					Chart for Medians
	Factor for Central Line	Factors for Control Limits				
		d_2	D_3	D_4	D_5	
n						A_3
2	1.128	0	3.267	0	3.865	2.224
3	1.693	0	2.574	0	2.745	1.265
4	2.059	0	2.282	0	2.375	0.829
5	2.326	0	2.114	0	2.179	0.712
6	2.534	0	2.004	0	2.055	0.562
7	2.704	0.076	1.924	0.078	1.967	0.520
8	2.847	0.136	1.864	0.139	1.901	0.441

Process Capability Ratio is relevant to X bar control chart, So the upper control limit (UCL) and lower control limit (LCL) from X bar control chart will be applicable for calculating C_p and C_{pk}

So, $\sigma = 9.857/2.059 = 4.787$ (For $n=4$, d_2 is equal to 2.059)

Process Capability Ratio (PCR) = $C_p = (\text{USL} - \text{LSL}) / 6\sigma$

$$= (48.6 - 34.22) / (6 \times 4.787) = 0.5$$

$$C_{pk} = \frac{\text{Min} \left\{ (USL - \bar{X}) \text{ or } (\bar{X} - LSL) \right\}}{3\sigma}$$

Here, USL-X bar = 48.6- 41.41=7.19

X bar-LSL=41.41-34.22=7.19

$$C_{pk} = 7.19 / (3 \times 4.787) = 0.5$$